Chapter 3. Resampling: Cross-Validation and the Bootstrap

Lecture 5 (validation set, LOOCV, K-fold CV, right and wrong CV, bootstrap SE and CI)

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1 What Are Resampling Methods?

Resampling methods repeatedly draw samples from a training set and refit the model of interest on each sample, to learn more about the fitted model:

- **model assessment**: estimate test error rates;
- **model selection**: choose the appropriate level of flexibility.

They are computationally expensive, but feasible with modern computers. Two methods: cross-validation and the bootstrap.

The best way to estimate prediction error is a large designated test set, which is often unavailable. AIC and BIC adjust the training error mathematically (Ch. 2). Here we instead hold out a subset of the training observations from fitting and evaluate on it.

2 Cross-Validation

2.1 Validation-set approach

Randomly split the samples into a training set and a validation set; fit on the training set, predict the validation set, and use the validation error (MSE for quantitative, misclassification rate for qualitative responses) as an estimate of the test error.

Limitations. (i) The estimate can be highly variable, depending on which observations fall in which set.

(ii) Only part of the data is used for fitting, so the validation error tends to *overestimate* the test error of the model fitted on all data.

2.2 Leave-one-out cross-validation

For each model: split the n observations into a training set of size $n-1$ and a validation set of size 1; fit, compute $\mathrm{MSE}_i = (y_i - \hat{y}_{(i)})^2$; repeat n times and average,

$$\mathrm{CV}_{(n)} = \frac{1}{n}\sum_{i=1}^{n} \mathrm{MSE}_i.$$

LOOCV has less bias than the validation-set approach (each fit uses almost all data) and no randomness in the split, but it requires n fits.

Theorem: LOOCV shortcut for least squares and linear smoothers

If $\hat{y} = Hy$ for a fixed matrix H with the leave-one-out property (true for least squares, ridge, smoothing splines), then

$$\mathrm{CV}_{(n)} = \frac{1}{n}\sum_{i=1}^{n} \left(\frac{y_i - \hat{y}_i}{1 - h_{ii}}\right)^2,$$

so LOOCV costs a single fit.

Proof for least squares [Supplement]. By the Sherman–Morrison formula, removing row i changes the estimate by $\hat{\beta} - \hat{\beta}_{(i)} = (X^\top X)^{-1} x_i e_i/(1-h_{ii})$ with $e_i = y_i - \hat{y}_i$. Hence $y_i - x_i^\top \hat{\beta}_{(i)} = e_i + h_{ii}e_i/(1-h_{ii}) = e_i/(1-h_{ii})$. ■

Points with high leverage h_{ii} have their residuals inflated, as they should be.

2.3 K-fold cross-validation

Randomly divide the data into K (roughly) equal parts $C_1, \ldots, C_K$ with n_k observations in part k ($n_k = n/K$ if K divides n). Leave out part k, fit on the remaining $K-1$ parts, and predict part k; do this for every k and combine:

$$\mathrm{CV}_{(K)} = \sum_{k=1}^{K} \frac{n_k}{n} \mathrm{MSE}_k, \qquad \mathrm{MSE}_k = \frac{1}{n_k} \sum_{i \in C_k} (y_i - \hat{y}_i)^2,$$

where $\hat{y}_i$ is fitted from the data with part k removed. $K = n$ gives LOOCV. The estimate is used both to select the best model and to give an idea of the test error of the chosen model.

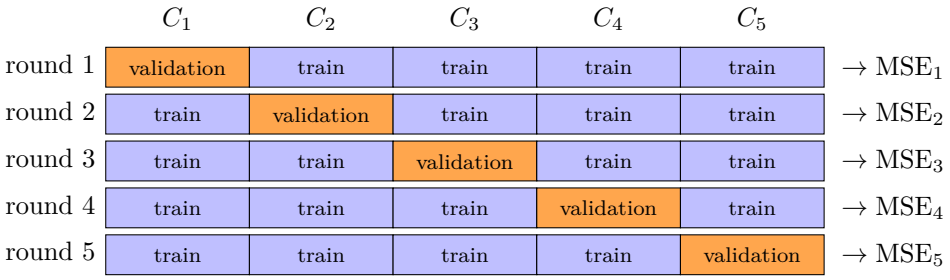


Figure 1: 5-fold cross-validation. Each part is held out once; the CV estimate averages $\mathrm{MSE}_1, \ldots, \mathrm{MSE}_5$. LOOCV is the extreme case with n parts of size one.

2.4 K-fold CV versus LOOCV

Typical choices are $K = 5$ or 10. The obvious advantage of K-fold CV is computational. Less obvious: LOOCV "does not shake up the data enough" — the n fitted models are trained on nearly identical data, so their errors are *highly correlated* and their average can have high variance. K-fold CV has somewhat more bias (training sets of size $(K-1)n/K$) but lower variance: a bias–variance trade-off for the error estimate itself. On the Auto data (mpg versus polynomials in horsepower), the LOOCV curve and nine repetitions of 10-fold CV with different random splits give very similar curves.

2.5 Cross-validation: right and wrong

Example: selecting predictors before CV

With 5000 predictors and 200 samples:
(1) select the 100 predictors most correlated with the response;
(2) fit a model using only these 100. Can we apply CV only to step 2? **No.** Step 1 has already used the responses of the entire data set, so the validation folds are not independent of the selection. Even if the predictors are pure noise, the selected 100 look predictive on all folds and the CV error is optimistically biased (in the classic simulation it is near 0 although the true error is 50%). The right way applies CV to *steps 1 and 2 together*: in each fold, select the predictors using only the training part.

Feature selection is a form of training and must be included in the validation process.

2.6 What does CV estimate?

For a model $\hat{f}$ trained on data, the prediction risk is $R(\hat{f}) = \mathbb{E}[(Y - \hat{f}(X))^2]$; new data are unavailable, so CV approximates the expected prediction error by repeatedly splitting into training and

validation parts. **CV estimates the prediction risk.**

Theory and limitation. Stone (1977) showed that LOOCV is asymptotically equivalent to AIC; both aim to minimize prediction risk, which justifies CV for prediction. But CV does not necessarily identify the true model: Shao (1997) showed that LOOCV may prefer overly complex models (it is not consistent for model selection, like AIC). Takeaway: prediction → CV, AIC; true-model identification → BIC.

3 The Bootstrap

The bootstrap is a flexible and powerful tool to quantify the uncertainty of an estimator or a learning method, e.g. the standard error of a coefficient or a confidence interval.

3.1 A motivating example

We invest a fraction α of our money in an asset with return X and $1-\alpha$ in an asset with return Y, and want to minimize the risk $\mathrm{Var}(\alpha X + (1-\alpha)Y)$.

Theorem: minimum-variance allocation

With $\sigma_X^2 = \mathrm{Var}X$, $\sigma_Y^2 = \mathrm{Var}Y$, $\sigma_{XY} = \mathrm{Cov}(X,Y)$,

$$\alpha = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}.$$

Proof [Supplement]. $V(\alpha) = \alpha^2\sigma_X^2 + (1-\alpha)^2\sigma_Y^2 + 2\alpha(1-\alpha)\sigma_{XY}$ is a convex quadratic; $V'(\alpha) = 2\alpha\sigma_X^2 - 2(1-\alpha)\sigma_Y^2 + 2(1-2\alpha)\sigma_{XY} = 0$ gives $\alpha(\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}) = \sigma_Y^2 - \sigma_{XY}$. ■

The variances are unknown, so we plug in estimates:

$$\hat{\alpha} = \frac{\hat{\sigma}_Y^2 - \hat{\sigma}_{XY}}{\hat{\sigma}_X^2 + \hat{\sigma}_Y^2 - 2\hat{\sigma}_{XY}}.$$

How accurate is $\hat{\alpha}$? In a simulation (true $\alpha = 0.6$), four data sets of 100 pairs gave $\hat{\alpha} = 0.576, 0.532, 0.657, 0.651$. Repeating 1,000 times gave $\bar{\alpha} = \frac{1}{1000}\sum_r \hat{\alpha}_r = 0.5996$ and standard deviation $\sqrt{\frac{1}{999}\sum_r(\hat{\alpha}_r - \bar{\alpha})^2} = 0.083$, so $\mathrm{SE}(\hat{\alpha}) \approx 0.083$.

3.2 Back to the real world

For real data we cannot draw new samples from the population. The bootstrap uses the computer to *mimic* this: instead of independent data sets from the population, it repeatedly samples observations **from the original data set with replacement**, each bootstrap data set having the same size n. Some observations appear several times, others not at all. In the portfolio example, histograms of 1,000 estimates from true resampling and from 1,000 bootstrap samples of one data set look very similar.

Algorithm: bootstrap standard error

For $b = 1, \ldots, B$ (say $B = 100$ or 1000): draw Z^{*b} by sampling n observations with replacement, and compute $\hat{\alpha}^{*b}$. Then

$$\mathrm{SE}_B(\hat{\alpha}) = \sqrt{\frac{1}{B-1}\sum_{r=1}^{B}\left(\hat{\alpha}^{*r} - \frac{1}{B}\sum_{r'=1}^{B}\hat{\alpha}^{*r'}\right)^2}$$

estimates the standard error of $\hat{\alpha}$ computed from the original data.

Why it works. The bootstrap replaces the unknown P_0 by the empirical distribution $\hat{P}_n$ (mass $1/n$ at each observation) — the *plug-in principle*. Sampling from $\hat{P}_n$ mimics sampling from P_0, and for smooth statistics the bootstrap distribution of $\hat{\alpha}^* - \hat{\alpha}$ approximates that of $\hat{\alpha} - \alpha$.

3.3 Bootstrap in general

More complex data need adapted schemes. For time series we cannot resample observations independently; the **block bootstrap** resamples blocks of consecutive observations with replacement and pastes them together, preserving short-range dependence.

3.4 Confidence intervals

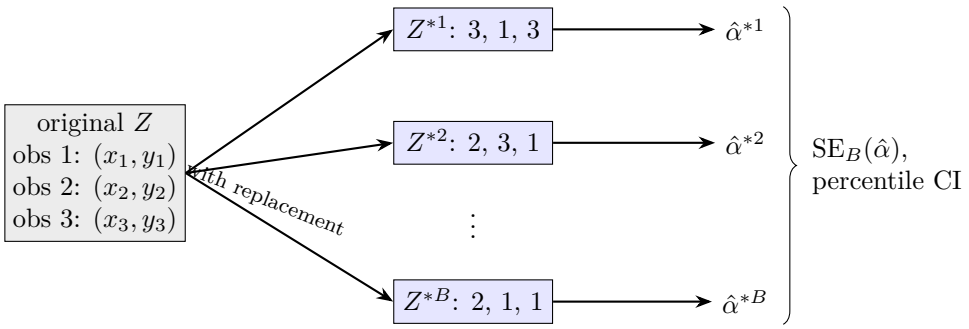


Figure 2: The bootstrap with $n = 3$. Each bootstrap set has n observations drawn with replacement (some repeated, some missing); the spread of the B estimates approximates the sampling distribution of $\hat{\alpha}$.

Besides standard errors, the bootstrap gives approximate confidence intervals. From the histogram of 1,000 bootstrap $\hat{\alpha}^*$, the 5% and 95% quantiles are $(0.43, 0.72)$: an approximate 90% **bootstrap percentile interval** for α. Alternatives: the normal interval $\hat{\alpha} \pm z_{1-\alpha/2}\mathrm{SE}_B$, and the basic interval $(2\hat{\alpha} - q_{0.95},\ 2\hat{\alpha} - q_{0.05})$.

3.5 The wrong way: bootstrap prediction error

One might use each bootstrap sample as a training set and the original sample as the validation set. In CV, validation folds are disjoint from training folds; but each bootstrap sample overlaps heavily with the original data: an observation appears in a given bootstrap sample with probability $1 - (1 - 1/n)^n \approx 1 - e^{-1} \approx 0.632$. This seriously *underestimates* the true prediction error. (Using only out-of-bag observations for each bootstrap fit fixes this, leading to the ".632" estimators and to OOB error in bagging, Ch. 11.)

Summary

- Resampling reuses the observed data for two goals: estimating prediction error and quantifying the uncertainty of an estimator.
- The validation-set estimate is variable and biased upward; LOOCV is nearly unbiased but its fold errors are highly correlated; $K = 5$ or 10 balances both.
- Every data-dependent step (including variable selection) must be inside the CV loop.
- CV is asymptotically like AIC (prediction), not like BIC (identification).
- The bootstrap resamples with replacement to estimate standard errors and confidence intervals; it is not a substitute for CV in estimating prediction error.
- Prediction → cross-validation; uncertainty → bootstrap.

4 Selected exercises

Motivation. The first item is ISLR Exercise 5.2 on the 0.632 inclusion probability; the other two prove the LOOCV shortcut for ridge regression and compute the bootstrap standard error of a mean exactly.

Exercise 3.1 (Inclusion in a bootstrap sample; ISLR Ex. 5.2)

A bootstrap sample draws n observations with replacement from n observations.
(a) What is the probability that the first bootstrap observation is not the jth original observation? The second?
(b) Show that the probability that observation j is not in the bootstrap sample is $(1-1/n)^n$, and find the expected number of distinct original observations in a bootstrap sample.
(c) Evaluate the inclusion probability for $n = 5, 100, 10\,000$, and prove that it decreases strictly to $1 - e^{-1}$ as $n \to \infty$ ($n \geq 2$).

Solution.

(a) Each draw is uniform on $\{1, \dots, n\}$, so the first draw differs from j with probability $1 - 1/n$; the same holds for the second draw.

(b) The n draws are independent, so $\Pr(j \notin \text{sample}) = (1 - 1/n)^n$. With I_j the indicator that j appears, the number of distinct observations is $\sum_j I_j$, with expectation $n\{1 - (1 - 1/n)^n\}$ by linearity.

(c) The values $1 - (1 - 1/n)^n$ are 0.672, 0.634 and 0.632.

Step 1. Write $(1 - 1/n)^n = e^{a_n}$ with $a_n = \varphi(1/n)$, where $\varphi(x) = \log(1 - x)/x$ for $0 < x < 1$. From $\log(1 - x) = -\sum_{k \geq 1} x^k/k$,

$$\varphi(x) = -\Big(1 + \frac{x}{2} + \frac{x^2}{3} + \cdots\Big),$$

which is strictly decreasing in x with $\varphi(x) \to -1$ as $x \downarrow 0$.

Step 2. As n increases, $x = 1/n$ decreases, so a_n increases strictly to -1, and $(1 - 1/n)^n$ increases strictly to e^{-1}. Hence the inclusion probability decreases strictly to $1 - e^{-1} \approx 0.632$: a bootstrap sample contains about 63% of the distinct observations, which is why it overlaps so much with the original data. ■

Exercise 3.2 (LOOCV shortcut for ridge regression)

Let $X \in \mathbb{R}^{n\times p}$ (rows $x_i^\top$, no intercept), $\lambda > 0$, $A = X^\top X + \lambda I$, $\hat\beta = A^{-1}X^\top y$, $\hat y = X\hat\beta$, and $h_{ii} = x_i^\top A^{-1} x_i$. Let $\hat\beta_{(i)}$ be the ridge estimate with the same λ computed without observation i.
(a) Prove the Sherman–Morrison formula: if A is invertible and $c = 1 - u^\top A^{-1}u \neq 0$, then $(A - uu^\top)^{-1} = A^{-1} + A^{-1}uu^\top A^{-1}/c$.
(b) Show that $h_{ii} < 1$ and $y_i - x_i^\top \hat\beta_{(i)} = (y_i - \hat y_i)/(1 - h_{ii})$.

Solution [Supplement].

(a) Multiplying out,

$$(A - uu^\top)\Big(A^{-1} + \frac{A^{-1}uu^\top A^{-1}}{c}\Big) = I + \frac{uu^\top A^{-1}}{c} - uu^\top A^{-1} - \frac{u(u^\top A^{-1}u)u^\top A^{-1}}{c}$$
$$= I + uu^\top A^{-1}\,\frac{1 - c - u^\top A^{-1}u}{c} = I.$$

(b)

Step 1 ($h_{ii} < 1$). If $x_i = 0$ then $h_{ii} = 0$. Otherwise put $v = A^{-1}x_i \neq 0$. For every w, $w^\top A w = \|Xw\|^2 + \lambda\|w\|^2 \geq (x_i^\top w)^2 + \lambda\|w\|^2$. With $w = v$: $v^\top A v = x_i^\top A^{-1}x_i = h_{ii}$ and $x_i^\top v = h_{ii}$, so $h_{ii} \geq h_{ii}^2 + \lambda\|v\|^2 > h_{ii}^2$, i.e. $0 \leq h_{ii} < 1$.

Step 2. Deleting row i gives $A_{(i)} = A - x_i x_i^\top$ and $\hat\beta_{(i)} = A_{(i)}^{-1}(X^\top y - x_i y_i)$. By (a) with $u = x_i$ and $c = 1 - h_{ii} > 0$,

$$x_i^\top A_{(i)}^{-1} = x_i^\top A^{-1} + \frac{h_{ii}\, x_i^\top A^{-1}}{1 - h_{ii}} = \frac{x_i^\top A^{-1}}{1 - h_{ii}}.$$

Step 3. Therefore $x_i^\top \hat\beta_{(i)} = \dfrac{x_i^\top A^{-1}X^\top y - h_{ii}y_i}{1 - h_{ii}} = \dfrac{\hat y_i - h_{ii}y_i}{1 - h_{ii}}$, and

$$y_i - x_i^\top \hat\beta_{(i)} = \frac{(1 - h_{ii})y_i - \hat y_i + h_{ii}y_i}{1 - h_{ii}} = \frac{y_i - \hat y_i}{1 - h_{ii}}.$$

So one ridge fit gives $\mathrm{CV}_{(n)}$ for each λ. With $\lambda = 0$ the same computation is the least-squares case, provided $A_{(i)}$ is invertible. (If the data are re-centered and re-standardized inside each fold, the identity holds only approximately.) ■

Exercise 3.3 (Bootstrap standard error of the mean)

Fix data $x_1, \ldots, x_n$ and let $x_1^*, \ldots, x_n^*$ be drawn i.i.d. from the empirical distribution $\hat P_n$, with mean $\bar x^*$.
(a) Show that $\mathbb{E}^*\bar x^* = \bar x$ and $\mathrm{Var}^*\bar x^* = \hat\sigma^2/n$, where $\hat\sigma^2 = \frac{1}{n}\sum_i (x_i - \bar x)^2$.
(b) Show that $\mathrm{SE}_B \to \hat\sigma/\sqrt{n} = \sqrt{(n-1)/n}\; s/\sqrt{n}$ almost surely as $B \to \infty$, with s^2 the usual sample variance.

Solution [Supplement].

(a) Each x_j^* takes the value x_i with probability $1/n$, so $\mathbb{E}^* x_j^* = \bar x$ and $\mathrm{Var}^* x_j^* = \frac{1}{n}\sum_i (x_i - \bar x)^2 = \hat\sigma^2$. The mean of n i.i.d. copies has mean $\bar x$ and variance $\hat\sigma^2/n$.

(b) Conditionally on the data, the bootstrap replicates $\bar x^{*1}, \ldots, \bar x^{*B}$ are i.i.d. and bounded. By the strong law of large numbers, $\frac{1}{B}\sum_r \bar x^{*r} \to \bar x$ and $\frac{1}{B}\sum_r (\bar x^{*r})^2 \to \hat\sigma^2/n + \bar x^2$ almost surely, so

$$\mathrm{SE}_B^2 = \frac{B}{B-1}\Big[\frac{1}{B}\sum_r (\bar x^{*r})^2 - \Big(\frac{1}{B}\sum_r \bar x^{*r}\Big)^2\Big] \to \frac{\hat\sigma^2}{n}.$$

Since $\hat{\sigma}^2 = (n-1)s^2/n$, the ideal bootstrap SE equals the textbook $s/\sqrt{n}$ up to the factor $\sqrt{(n-1)/n} \to 1$: for the mean, the bootstrap reproduces the classical answer, and its value lies in statistics without such formulas, like $\hat{\alpha}$. ■